INCH-POUND

MIL-PRF-1/839F
3 October 2005
SUPERSEDING
MIL-E-1/839E
9 June 1981

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AN5WA

Inactive for new design
after 2 October 2000

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Pentode, miniature, video amplifier

Outline - - - - - - - - - 5-2 (EIA)
Base - - - - - - - - - - E7-1
Envelope - - - - - - - T5-1/2
Cathode - - - - - - - - Coated unipotential
Base connections:

Pin No. - - - - - - - -	1	2	3	4	5	6	7
Element - - - - - - -	g1	k,g3	h	h	a	g2	k,g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rk	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	V	V dc	V dc	V dc	V	Ohms	MegΩ	mA dc	W	W	°C	ft
Maximum:	6.9	135	0, -55	135	200	---	0.1	55	4.6	1.55	+200	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	120	0	120	0	125	---	---	---	---	---	---

GENERAL:

Qualification: Not required.

Reliable tube.

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TABLE I. Requirements.

Method of MIL-STD-1311	Requirement or test	Notes	Conditions	Acceptance level (see note 10)	Symbol	Limits		Unit
						Min	Max	
	Conformance inspection, part 1							
1301	Heater current			0.4	If	420	480	mA
1336	Heater-cathode leakage			0.4	Ihk	---	20	μA dc
1266	Total grid current	2		0.4	Ic1	0	-2.0	μA dc
1256	Electrode current (1) (anode)	2		0.4	Ib	25	43	mA dc
1341	Power output		Esig = 4.25 V ac; RL = 2,500 ohms	0.4	Po	1.0	---	W
1201	Short and discontinuity detection			0.4	---	---	---	---
	Conformance inspection, part 2							
1211	Insulation of electrodes			2.5	R	---	---	---
1256	Electrode current (1) (screen grid)			2.5	Ic2	6.0	16.0	mA dc
1256	Electrode current (2) (anode)		Ec1 = -20 V dc	2.5	Ib	---	1.0	mA dc
1306	Transconductance (1)			2.5	Sm	7,000	10,000	μmhos
1266	Grid emission	3	Ef = 7.5 V; Ec1 = -45 V dc; Rg1 = 0.01 MΩ (min)	2.5	Ic1	0	-4.0	μA dc
1306	Transconductance (2)		Ef = 5.7 V	2.5	$\frac{\Delta Sm}{Ef}$	---	10	%
1231	Emission		Eb = Ec2 = Ec1 = 15 V dc	2.5	Is	100	---	mA dc
2201	Noise and microphonics		Esig = 500 mV ac Rp = 2,000 ohms Rg1 = 0.1 MΩ	2.5	---	---	---	---

See footnotes at end of table.

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TABLE I. Requirements - Continued.

Method of MIL-STD-1311	Requirement or test	Notes	Conditions	Acceptance level (see note 10)	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 2 - Continued							
1331	Direct-interelectrode capacitance		Shield No. 316 Shield No. 316 Shield No. 316	6.5	Cglp Cin Cout	--- 6.0 4.0	0.075 12.0 7.0	pF pF pF
1256	Electrode current (3) (anode)		Eb = Ec2 = 60 V dc; Ec1 = 0; Rk = 0	2.5	Ib	25	---	mA dc
1256	Electrode current (2) (screen grid)		Eb = Ec2 = 60 V dc; Ec1 = 0; Rk = 0	6.5	Ic2	6.5	15.5	mA dc
1031	High-frequency vibration		Rp = 2,000 ohms	2.5	Ep	---	150	mV ac
1031	Vibration-fatigue	4		6.5	---	---	---	---
1041	Shock	4, 5	450 G; Ehk =100 V dc	---	---	---	---	---
---	Shock and vibration-fatigue-test end points:							
1031	High-frequency vibration		Rp = 2,000 ohms	---	Ep	---	300	mV ac
1336	Heater-cathode leakage			---	Ihk	---	50	µA dc
1266	Total grid current			---	Ic1	0	-4.0	µA dc
1341	Power output		Esig = 4.25 Vac; RL = 2,500 ohms	---	Po	0.8	---	W
1121	Base strain	6		---	---	---	---	---
2126	Envelope strain			2.5	---	---	---	---
1105	Permanence of marking			---	---	---	---	---

See footnotes at end of table.

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TABLE I. Requirements - Continued.

Method of MIL-STD-1311	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = +100 V dc; Ec1 = Ec2 = Eb = 0; 1 min "on", 4 min "off"	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	30	µA dc
1516	Stability life		Ehk = +135 V dc; Rg1 = 0.1 MΩ TA = room temperature	---	---	---	---
---	Stability life-test end point (2 and 20 hours):						
1306	Change in trans-conductance (1) of individual tubes			ΔSmt	---	10	%
1501	Intermittent life	8	Ehk = +135 V dc; Rg1 = 0.1 MΩ TE = 200°C (min)	---	---	---	---
---	Intermittent life-test end points (1,000 hours):						
1266	Total grid current			Ic	0	-3.0	µA dc
1301	Heater current			If	410	490	mA
1306	Change in transcon-ductance (1) of individual tubes			ΔSmt	---	20	%
1341	Power output		Esig = 4.25 V ac; RL = 2,500 ohms	Po	0.50	---	W
1336	Heater-cathode leakage			Ihk	---	50	µA dc
1211	Insulation of electrodes			R	50	---	MΩ
1511	Cathode interface life	7	Ef = 6.9 V; Ehk = 0	Ri	---	25.0	Ohms

See footnotes at top of next page.

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TABLE I. Requirements - Continued.

NOTES:

1. See "Reduced pressure (altitude) rating," and altitude, maximum peak voltage in the basic document.

2. This test shall be performed at the conclusion of the holding period.

3. Prior to this test, the tubes under test (TUT) shall be preheated a minimum of 5 minutes operating at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Ec1	Eb	Rk/k	Rg/g	Ec2
V	Vdc	Vdc	Ohms	MegΩ	Vdc
7.5	0	120	125	0.01	120

4. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes.

5. A grid resistor of 0.1 megohm shall be added except when a thyratron-type short indicator is used.

6. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (see MIL-STD-1311, method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   a. Accepted if not one defective for class "A", "B", or "C" defects, respectively (see MIL-STD-1311, method 1121), or if not one total defective is found in the sample.

   b. Rejected if one or more defectives for class "A", "B", or "C" defects, respectively, or if a total of one defective or more is found in the sample.

7. The life-test sample shall consist of 20 tubes per lot and no tube failure shall be permitted. In the event of rejection of the first sample due to failure, a second sample of 20 tubes shall be selected from the lot. Acceptance shall then be based on the second samples, with no failures allowed.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (see MIL-STD-1311, method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life test-rack.

9. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

10. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

11. Referenced documents: In addition to MIL-PRF-1, this document references the following: MIL-STD-1311.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA-CC

(Project 5960-2005-002)

Review activities:
Army - AR, MI
Navy - AS, CG, OS
Air Force - 99

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